Q1 · Dataset and setup

What does the data look like, and how do we use it?

WHAT WE DID

- Dropped pairs without a label (−1)
- Random sample of **50k** training pairs

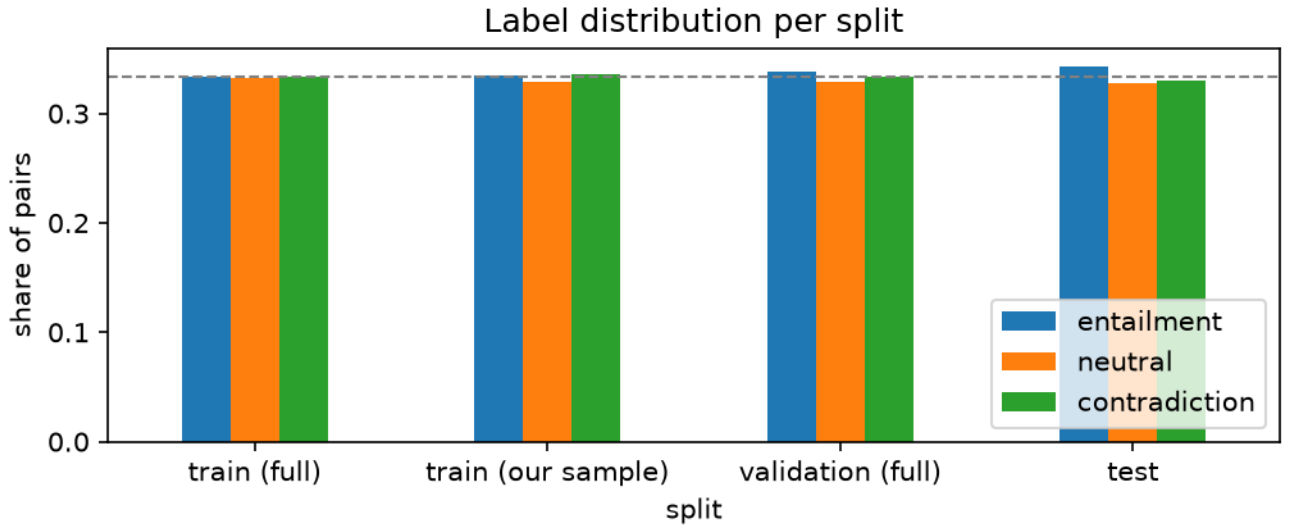
WHY

- −1: annotators disagreed
- 50k: 9 x 5 = 45 CV runs stay feasible

WHAT WE FOUND

- Labels ≈ 1/3 in every split
- Majority baseline **0.336**

P *A man holding a video camera standing next to a woman by water.*
H *The man is holding a microphone as he introduces the woman standing by the water.*
→ **contradiction**



Decision → accuracy as metric · 50k for CV · 2k validation pairs for design · test set once

Q2 · Tokenizer

Can we reuse the tutorial's tokenizer, and is our data enough to train a new one?

WHAT WE DID

- BPE on news vs SNLI text
- Several vocabulary sizes
- min_frequency = 2: BPE only merges a pair of tokens if it occurs at least twice.

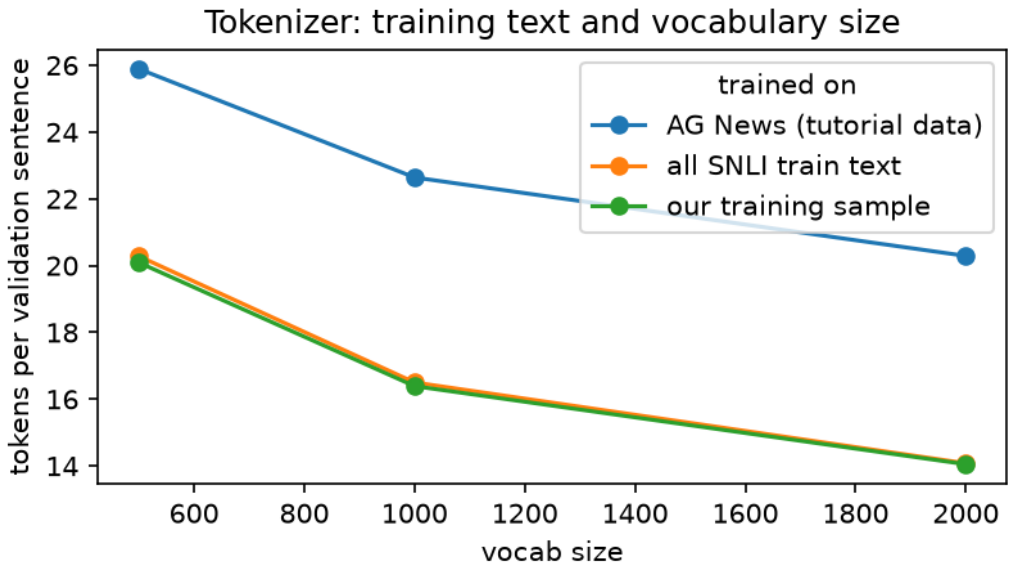
WHY

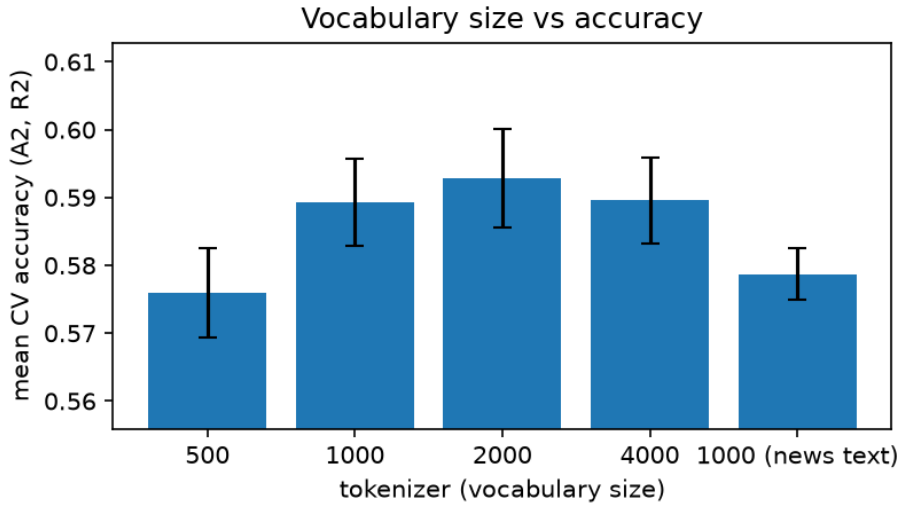
- SNLI = short captions, not news

WHAT WE FOUND

- News: **22.6** vs **16.5** tokens per sentence
- Our 50k sample ≈ all SNLI
- 1000 ≈ 2000 in accuracy

Decision → own BPE tokenizer on SNLI training text, vocabulary 1000





https://github.com/vidalisa/natural-language-inference.git

Q3 · Representations

How do we turn a premise–hypothesis pair into features?

WHAT WE DID

- **R1** 0/1 · **R2** counts, P and H side by side
- **R3** = R2 + cross-sentence features

WHY

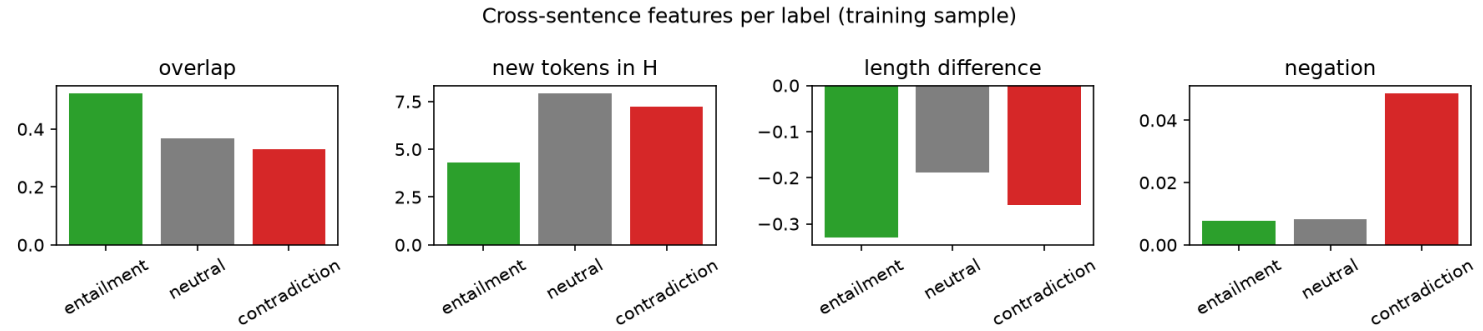
- NLI is about how P and H relate
- A linear model can't compute that

WHAT WE FOUND (accuracy on val)

- R1 ≈ R2
- R3 **+0.079** for every model



R1, R2: first two blocks (2,000 dims) · R3: all four (3,003 dims)



Training sample: each new feature separates the labels before any training

Decision → R3; R1 vs R2 makes no practical difference

Q4 · Architectures

Which models do we compare, and how did we design the third one?

WHAT WE DID

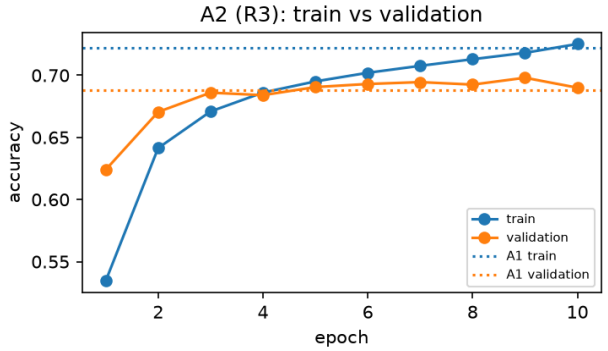
- **A1** log-linear · **A2** MLP (tutorial)
- **A3**: 2nd hidden layer, tuned

WHY

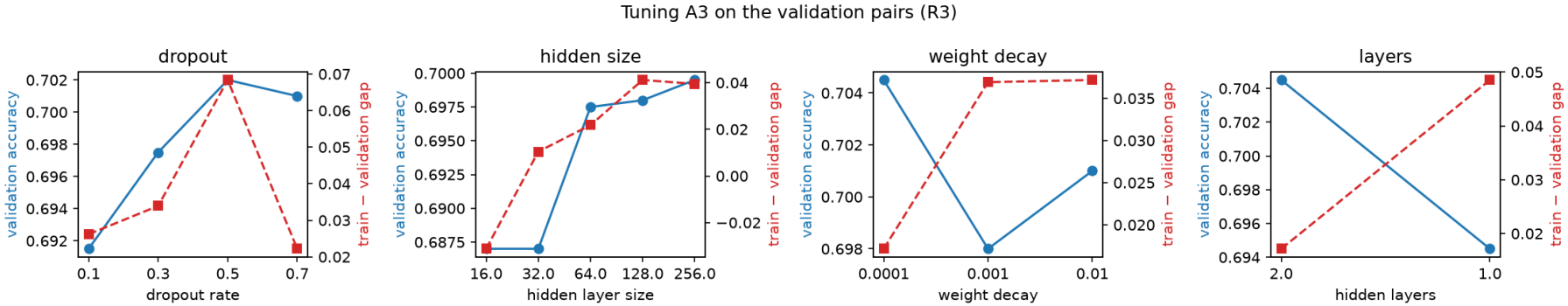
- A1, A2 barely overfit → add capacity

WHAT WE FOUND

- **A1 ≈ A2 ≈ A3** on R3



A2: validation flattens after ~5 epochs



A3 tuning, left → right · blue: validation accuracy · red: train–val gap

Decision → the representation matters more than the model

Q5 · Evaluation

How do we measure performance, and is that adequate for NLI?

WHAT WE DID

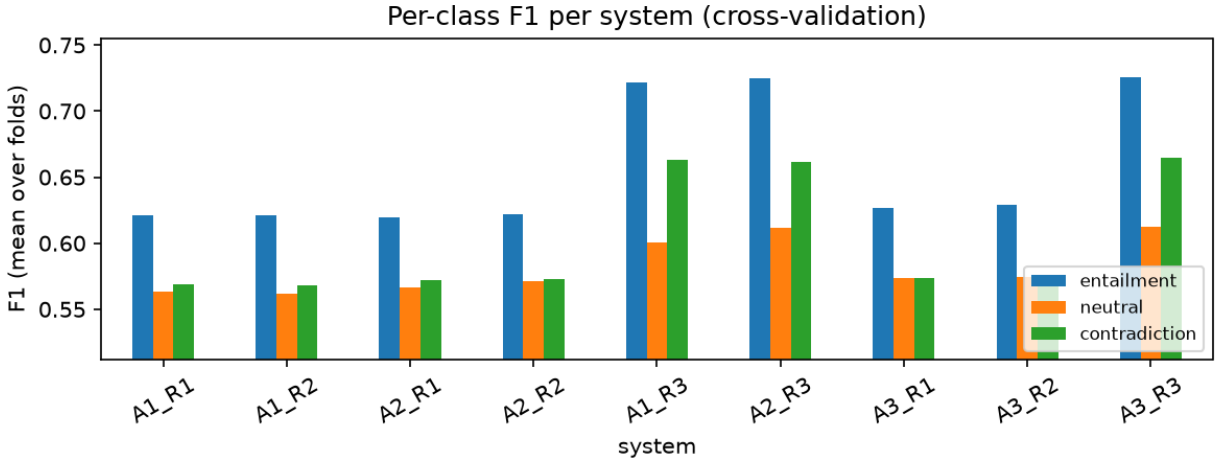
- 5-fold CV, same folds for all
- Accuracy + per-class F1 (how well each label is found and predicted)

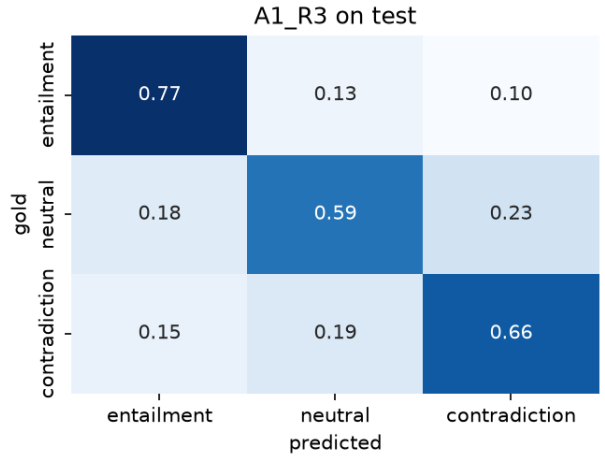
WHY

- Balanced labels → accuracy is fair
- F1 shows which label fails

WHAT WE FOUND

- **Neutral** is hardest for every system
- Macro-F1 gives the same ranking





Decision → compare on CV accuracy; use F1 to explain the errors

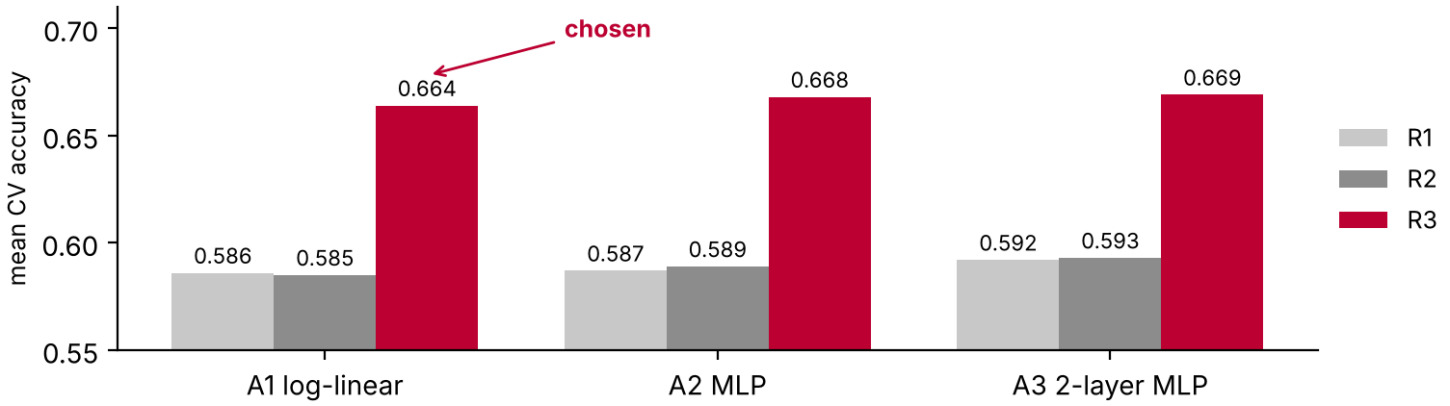
https://github.com/vidalisa/natural-language-inference.git

Q6 · Choosing the best system

Which system is best, and how sure are we?

WHAT WE DID

- Bayesian estimation on 5 fold accuracies: how likely is each possible difference? *(distribution of differences)*
- ROPE: differences less than 0.01 count as equal



WHY

- Simplest system that is equivalent to the best

WHAT WE FOUND

- A1_R3 ≈ A3_R3 (best)
- Test: accuracy **0.675** · macro-F1 **0.673**

comparison	difference	P(equivalent)	verdict
R1 → R2 (A1)	−0.001	0.997	equivalent
R2 → R3 (A1)	+0.079	0.000	R3 better
A1 → A2 (R3)	+0.004	0.983	equivalent
A2 → A3 (R3)	+0.001	1.000	equivalent

P(equivalent): area under bayesian estimation curve inside the ROPE band

Decision → A1_R3: the simplest system that is as good as the best